

Bulk XML Import

Load employee XML files into a normalised SQL Server HR database

SQL Server | T-SQL stored procedures | XML shredding | Data modelling

T-SQL

XML / XQuery

ETL

SQL Server 2016+

22

database tables

2

stored procedures

10

sample XML files

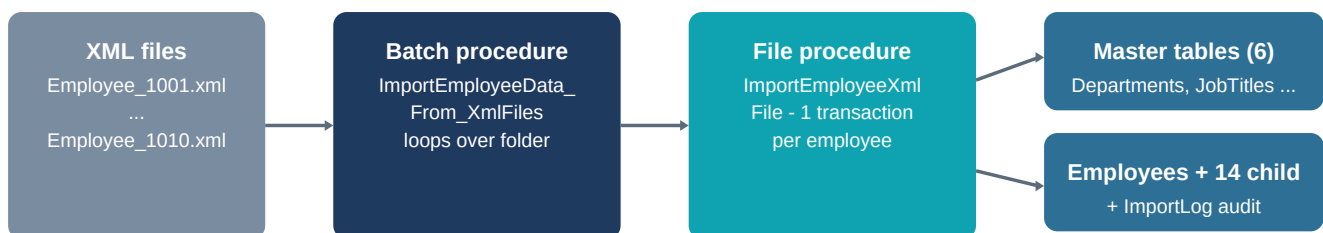
430

rows loaded from samples

Project overview

A production-style ETL routine written entirely in T-SQL. It reads employee records from XML files, shreds them with XQuery (`nodes()` / `value()`) and loads a normalised HR schema: an **Employees** table linked to **Departments**, **Job Titles**, **Locations** and other master tables, plus 14 child tables for addresses, skills, leave, salary history, timesheets and more. The design follows a proven patient-records import pattern and improves on it with transactions, audit logging and automatic master-data handling.

How it works



Master values are get-or-created; failures roll back that employee only and are recorded in ImportLog.

Key features

- **Safe by design** - one transaction per employee; any error rolls back that employee and is logged, the batch carries on.
- **Master data handled automatically** - if a file contains a department, job title, location, employment type, leave type or project that is not in the database yet, it is added automatically and linked to the right record (the employee, a leave entry or a timesheet entry).
- **Re-runnable** - re-importing a file replaces that employee cleanly using `ON DELETE CASCADE`.
- **Clean data handling** - empty tags become NULL (never 1900-01-01 or 0); UTF-8, accents and XML-escaped characters are supported.
- **Exact parent/child linking** - timesheet headers and entries are matched with `MERGE ... OUTPUT`, not row order.
- **No xp_cmdshell** - folder files are listed with `xp_dirtree`; every run is auditable in `ImportLog`.

Database schema

Group	Tables
Master / lookup (6)	Departments (self-referencing), JobTitles, Locations, EmploymentTypes, LeaveTypes, Projects
Parent (1)	Employees - FKs to Department, Job Title, Location, Employment Type; manager link; source file; import time
Personal	EmployeeAddresses, EmployeeEmergencyContacts, EmployeeDependents
Career	EmployeeQualifications, EmployeeWorkHistory, EmployeeSkills, EmployeeCertifications, EmployeePerformanceReviews, EmployeeTraining
Payroll and time	EmployeeSalaryHistory, EmployeeLeave, EmployeeTimesheets, TimesheetEntries (two-level parent/child)
Other	EmployeeAssets, ImportLog (audit), vw_EmployeeOverview (reporting view)

Run it in four steps

```

1. sql/01_Create_Employee_Tables.sql ..... creates EmployeeDB, tables, seed data
2. sql/02_ImportEmployeeXmlFile.sql ..... imports one file
3. sql/03_ImportEmployeeData_From_XmlFiles.sql .. imports a folder
4. EXEC dbo.ImportEmployeeData_From_XmlFiles
   @FolderPath = N'C:\BulkXmlImport\xml';

```

XML input to database

XML element	Target table(s)	XML element	Target table(s)
Demographics	Employees + masters	SalaryHistory	EmployeeSalaryHistory
Address	EmployeeAddresses	Leave	EmployeeLeave + LeaveTypes
EmergencyContact	EmployeeEmergencyContacts	PerformanceReview	EmployeePerformanceReviews
Dependent	EmployeeDependents	Training	EmployeeTraining
Qualification	EmployeeQualifications	Asset	EmployeeAssets
WorkHistory	EmployeeWorkHistory	Timesheet	EmployeeTimesheets
Skill / Certification	EmployeeSkills / ...Certifications	Timesheet > Entry	TimesheetEntries + Projects

Sample data and verification

Ten fictional employees across six countries exercise the edge cases: a terminated employee, an employee on leave, new departments and locations created on import, empty tags, missing sections, and special characters (O'Connor, José Álvarez, Zoë, &, <internal>). A verification script provides row counts, an employee overview and reports for headcount, expiring certifications, leave, project hours and reporting lines.

Deliverable	Description
4 SQL scripts	Schema + seed data, single-file import, folder import, run and verify
10 XML files + generator	Deterministic Python generator to recreate or extend the data set
README.md	Setup, XML format, ER diagram, parameters, troubleshooting, extension guide
This overview (PDF)	Portfolio summary for project catalogues

Available for: XML/JSON/CSV import pipelines, SQL Server stored procedures, data modelling and migration, reporting views and query tuning.